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Chapter 37: Agency Is All You Need—Safeguarding Human Agency through Meta-Audit (the Extension of Innate Knowledge; Re-Enlightenment)

Abstract

In 2017, Vaswani et al. published "Attention Is All You Need" (Vaswani et al., 2017); the Transformer architecture defined the foundational operation of contemporary artificial intelligence (AI) through the allocation of attention weights. To date (September 11, 2026), this paper has accumulated over 260,000 citations. This chapter argues: it is not enough.

Three concepts must be strictly distinguished. Attention is a clock-dimension operation of allocating weights among given readings—the core mechanism of the Transformer; meta-audit is the act of examining the relationship between readings and the real—the capacity to ask "is this clock reliable?", a human capacity that can be supported by engineering (the four engineering directions of Chapter 9, §7 activate the user's meta-audit); agency is the mode of existence of the human being, and that mode of existence is the continuous, multi-channel, relational time dimension. The three form a hierarchy from the clock dimension to the time dimension: attention allocates weights, meta-audit questions the premises of weight allocation, and agency spontaneously generates questions—questions that cannot be traced back to any pre-set question or conditions for action.

Current AI possesses clock-dimension attention and can be engineered to imitate human meta-audit, but its operation remains discrete, graduated, and reading-only.

The argument proceeds along four lines of inquiry. First (§1), the historical return of mapping function monopoly—the medieval Church monopolized salvation readings through language and theology (Layer One: an information deficit that could be reduced through Luther-style translation), while AI black boxes reconstruct the same monopoly through a parameter space of trillions of dimensions (Layer Three: ontological dimensionality reduction, which cannot be expected to be fully opened); the four engineering directions of Chapter 9, §7 are not about opening the black box, but about making the black box honestly annotate its own uncertainty. Second (§2), the civilizational skew of the second derivative of narrative load (ϕ) within the black box—the individual as the default unit of analysis (with individualism as the default framework), Western-centrism as the default calibration baseline, transmitted through infrastructure rather than persuasion, bypassing the constraints formed within civilizations through distributed moral judgment. Third (§3), the differences and shared limits of the three governance clocks of China, the U.S., and the EU—the U.S. emphasizes projection-side expansion, the EU procedural frameworks, and China targeted intervention; the strengths and blind spots of these orientations differ but also overlap, and the effects of combining them depend on review resources, implementation capacity, and correction procedures, while the ownership of the global governance master clock remains undetermined, which is itself a mapping self-reference. Fourth (§4), two archetypal strategies in flood myths—the Ark (discretization + system reboot, waiting for salvation) and water management (continuous channeling + thirteen years of sustained work, not presupposing salvation).

Final judgment (§5): **Civilization exists for the sake of human beings; human beings do not exist to serve civilization.** Meta-audit is the minimal operationalization of agency under current technical conditions—it is about ensuring AI does not eliminate human agency. As for the attempt to equate ten quadrillion clocks with time, it is mathematical self-propagation.

Keywords

Agency; meta-audit; attention mechanism; mapping function monopoly; narrative load second derivative; three governance clocks; Noah's Ark and Yu the Great's flood control; the Clock Room; von Neumann

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§0 The Distinction Between Agency and Meta-Audit

Before entering the main text, a precise distinction must be drawn between two core concepts that run throughout the chapter. This distinction is not a terminological preference—it marks two different epistemological levels.

Attention is the operation of allocating weights among readings. The self-attention mechanism of the Transformer does precisely this: given a token sequence, it computes the relevance weight of each token relative to the positions accessible to it, and assembles information accordingly. This is an operation within the clock dimension—within a given set of clock readings, deciding which clock to look at, with what weight, and how to combine the readings of multiple clocks. The entire premise of the operation is: the readings have already arrived.

Meta-audit is the act of examining the relationship between a reading and the real—not merely reading the clock, but asking "is this clock reliable?" Meta-audit requires a trigger condition (a reading has arrived) and a set of criteria (the verification tools provided by the legitimacy base ($\|L\|$)). It is operationalizable—the four engineering directions of Chapter 9, §7 (ensemble disagreement, conformal prediction sets, evidence provenance, and

friction architecture; Chapter 16 developed these as the four paths of forced inconsistency, calibration, grounding metrics, and friction architecture, hereafter referred to by their short names) all provide engineering support under different conditions for meta-audit: model disagreement offers points of comparison, calibration methods constrain uncertainty claims, provenance supports the examination of evidence, and friction preserves opportunities for review; whether they actually support meta-audit depends on whether the information is usable and tested, and whether the review can affect subsequent action. Systems can be designed to support meta-audit; whether meta-audit occurs can be measured. Meta-audit is an act. The discussion distinguishes an act of meta-audit from the capacity for meta-audit: the former is a particular examination; the latter is the capacity to initiate and complete such examinations. That capacity requires maintenance but is not thereby identical to agency itself.

Agency, by contrast, is not a capacity—it is the mode of existence of the human being. What Type Four of Chapter 1, §1.2 (the meta-information problem) measures is precisely this: a new clock is pushed into the room; no one asks a question, no conditions for action are given; only an event has occurred. To recognize the event itself as a signal that calls for explanation, and thereby to spontaneously generate a question—"Why was this new clock brought in? On what was it calibrated?"—is the mode of existence of the human being within the time dimension (Chapter 1, §5.2: the time dimension is the continuous, multi-channel, relational mode of existence; the clock dimension is the discrete, graduated, reading-only mode of operation). This criterion is a positive description of the human cognitive situation, not a test for machines: the question a person spontaneously asks cannot be traced back to any pre-set question or conditions for action.

All of AI's behavior, by contrast, occurs within the clock dimension—processing within an operational mode, whether triggered by an immediate instruction, a pre-set rule, or a random number. To read the appearance of "spontaneous questioning" as a sign of agency is to take operation for existence—the recurrence of taking readings for the real. Searle's Chinese Room distinguishes formal symbol manipulation from understanding and points in the same direction: completeness of syntax does not constitute semantics. In current architectures, AI's trigger surface (prompts, system workflows, objective functions) is a given engineering fact—agency does not occur in the structure of AI, while it is ready to hand in the existence of the human being. Meta-audit is therefore human: the "meta" presupposes a self answerable for its own judgments—the examiner and the judgment examined belong to one and the same self—the object of scrutiny may be one's own judgment or an external reading one is considering relying on; the reflexivity lies in the fact that reliance itself is a judgment, and must likewise be examined—the extension of innate knowledge speaks of exactly this structure. An AI system's self-inspection is processing, not meta-audit; it can be engineered to imitate human meta-audit, but it is not an instance of meta-audit.

The hierarchical relationship among the three:

Concept	Definition	Corresponds to Ch.1 §1.2	Temporal Nature	Does AI Possess It?
Attention	Allocating weights among given readings	Types One–Two: processing within readings	Clock Dimension	Yes—the core of the Transformer

Concept	Definition	Corresponds to Ch.1 §1.2	Temporal Nature	Does AI Possess It?
Meta-Audit	Examining the relationship between readings and the real	Type Three: judgment for action	Clock–Time Interface	No—can be engineered to imitate; meta-audit is human
Agency	The mode of existence of the human being: when no question has been given, recognizing the event itself as a signal that calls for explanation	Type Four: spontaneous questioning	Time Dimension	No—remains discrete, graduated, and reading-only

Attention is the most basic thing a person does—looking. Meta-audit is the higher-order thing a person does—questioning what one sees. Agency manifests in this: even without an explicit request, a person can still decide whether to attend and whether to question.

MacKenzie (2009), relaying Callon's concept of agencement, provides a complementary perspective—"For Callon, an actor is 'made up of human bodies but also of prostheses, tools, equipment, technical devices, algorithms, etc.'—in other words is made up of an agencement" (MacKenzie, 2009). Agencement describes how the capacity to act is assembled—human bodies plus tools plus algorithms plus procedures, assembled into a composite actor. This chapter asks how scrutiny is organized within this composite actor: has the audit dimension been preserved? If the audit component within the agencement is compressed to zero—tools substituting for judgment, algorithms substituting for calibration—the exercise of agency degenerates into the execution of given procedures: an actor that acts without knowing what it is doing, why it is doing it, or whether it is doing it right. The distance from agencement to agency is precisely the distance from "an assembled capacity to act" to "a subject aware of how its capacity to act has been assembled."

Von Neumann's three words from 1955 all lie at the level of agency. Patience is not an act of auditing—it is a state of existence: existing in time without being driven by any specific clock. One does not need to be triggered in order to be patient—patience is a state maintained even in the absence of any input. Flexibility is not switching between two methods of auditing—it is the capacity to switch between two modes of existence (the time dimension and the clock dimension). Intelligence is not "auditing the reading of some clock"—it is "seeing the structure of the clock itself." All three words describe agency, not audit.

The argumentative structure of this chapter is therefore: **What is needed is agency. But agency cannot be directly engineered—one cannot tell an engineer "please implement agency in the AI." Meta-audit is the minimal operable slice of agency under current technical conditions.** The four paths of Chapter 16, §5—forced inconsistency, calibration, grounding metrics, friction architecture—are not about implementing agency in AI, but about preserving operational space for human agency within the design of AI. It is not about giving AI agency—it is about ensuring AI does not eliminate human agency.

This distinction determines the direction of the entire chapter: not "how to make AI smarter"—but "how to protect the conditions of existence for human beings as subjects in a world of increasingly intelligent AI." Agency is what is needed. Meta-audit is what can be done. The former is the end. The latter is the means.

Meta-audit in this sense is **re-enlightenment**—not enlightenment in the sense of calibrating clocks with clocks: that version replaces old readings with new ones and instills "light" as a new narrative—a continuation of occlusion. Kant warned that "new prejudices will serve as well as old ones to harness the great unthinking masses" (Kant, 1784)—Kant's point is that political revolution does not automatically reform ways of thinking; this book extends the warning: a doctrine that once broke through old occlusions may itself become a new occlusion when it ceases to be open to scrutiny. Wang grounds this work in the original clarity of liangzhi: innate knowledge is "luminous, shining, and not beclouded" (Ivanhoe, 2009; cf. *Daxue Wen*)—the light is original; yet the ordinary mind alternates between clarity and obscuration, and occlusion returns as it is removed, hence the effort "seeks only to decrease day by day, never to increase" (Wang Yangming, *Chuanxilu*; author's translation). Wang Yangming called the work of de-occlusion the extension of innate knowledge: removing the occlusion in the midst of affairs to restore the original light of the mind (Wang Yangming, *Chuanxilu*; author's translation)—not a one-time bestowal of light, but sustained de-occlusion. All the paths of meta-audit stand on this "re": they add nothing to the person; they only move the occluding objects away, again and again.

§1 From Indulgences to the Black Box: The Historical Return of the Mapping Function Monopoly

§1.1 The Medieval Clock Room

The medieval Catholic Church constructed a complete Clock Room.

The real state (T) was the state of the soul—whether the faithful were saved. The mapping function (π) was the authority to interpret doctrine—the rule mapping the behavior of the faithful to the Church's judgmental readings (r); whether the state of salvation (T) changed accordingly is precisely what that mapping claimed, not what it verified. The reading (r) was the Church's judgment—"sinful" or "absolved."

The faithful could not verify independently. Not because the sun did not exist—the faithful believed God existed. It was because the window to the sun was institutionally monopolized. The Church claimed to be the sole legitimate intermediary—only priests trained in theology and authorized by the Church could interpret scripture, administer the sacraments, and convey the divine will. Scripture was written in Latin—ordinary believers, ignorant of Latin, could not read it themselves. Doctrine was argued by the Church's theological system—and that theological system derived its institutional authority from the papacy. This was the closed-loop pseudo-calibration of Chapter 1, §3.6: calibrating a clock with a clock, arguing doctrine with doctrine.

The position of the indulgence within this structure is precise: **paying money changes the reading (r)—whether it changes the real state (T) is not determined in this section.** The buyer's behavior did not change in the case analyzed here; what changed was merely the reading given by the Church: from "time in purgatory" to "time shortened." This is a pure operation of narrative load (ϕ)—the fiber was artificially manipulated. The Church was selling not salvation, but the reading of salvation.

The boundaries of this chapter must be declared immediately: the above analysis does not concern the truth or falsity of faith itself—that lies beyond the applicable domain of the Clock Room framework. Chapter 1, §3.4's

definition does not restrict the type of T (Chapter 9, §3 has already restricted its core scope to states of affairs that can be independently calibrated). Whether the state of the soul belongs to this set is a question for theology and philosophy, not for this chapter. What this chapter analyzes is the institutional monopoly structure of the mapping function—whatever T is, the operation of monopolizing the channel to T and monetizing access to that channel can be precisely described in the language of the Clock Room. This structure holds independently, without touching the content of faith.

§1.2 Gutenberg's Dual Tool

In the mid-fifteenth century, Gutenberg developed a metal movable-type printing system in Europe. Early printed products included Bibles and indulgences.

The same technology simultaneously served two directions. The Church used it to amplify narrative load (ϕ)—printing indulgences on a large scale, extending the previously hand-copied, small-scale-circulation indulgence operation into industrial production. Luther used it to open the window—in 1522, he translated the New Testament into German and reproduced it in large quantities using the printing press, enabling more ordinary people to read scripture for themselves, bypassing the priestly mapping function (π)—the German translation expanded ordinary readers' access to scripture and reduced their dependence on established interpretive intermediaries, while translation itself continued to involve interpretive choices.

This is the historical prototype of the adaptive deviation of Chapter 1, §4.2. The printing press (a massive release of unlocked stock (C)—transforming writing from a scarce, hand-copied product into a mass-producible commodity) was used by both sides in the competition: the monopolist used it to reinforce the monopoly; the challenger used it to break the monopoly. Technology itself does not determine the direction. What determines the direction is the power relationship between the user and the technology.

Luther's Reformation, within the framework of the Clock Room, was a successful First-Layer solution of Chapter 1, §4.1: information deficit was eliminated. Latin was a language barrier—translation eliminated the barrier. Scarcity of reproduction was the dissemination barrier—the printing press eliminated the barrier. Ordinary believers gained a channel, separate from the Church, for contact with scripture (the sun, in their view). The distribution of textual access and interpretive authority changed—what changed here was access and interpretive authority, not the independent verification of salvation.

§1.3 The AI Black Box: Mapping Function Monopoly under Conditions of Third-Layer Irreversibility

Contemporary AI commercialization has reconstructed the medieval Church's structure of mapping function monopoly—but at a deeper, more irreversible level.

Five Levels of Structural Isomorphism:

1. **Monopoly of the Mapping Function.** The Church monopolized the authority to interpret doctrine—only priests could interpret scripture. AI companies monopolize frontier training—only entities possessing computing power and data can train frontier models; some commercial models also restrict access to weights and training information. Chapter 9, §5 has already argued: training corpora are overwhelmingly dominated by English (roughly 44% of Common Crawl is English; Chinese roughly 4.6%), and tokenizers optimized on English-language corpora exhibit systematic efficiency discrimination against other languages. The capacity to produce readings is concentrated in the hands of a few entities.

2. **The Structural Level of Opacity.** In the Middle Ages, scripture was written in Latin—a language barrier. Model parameters are stored in a numerical space of trillions of dimensions—a dimensional barrier. The sources of opacity in the two cases differ, but the effect is the same: the user is structurally excluded from the mapping function. The key difference is: Latin could be translated (Luther did it)—this is the First-Layer problem of Chapter 1, §4.1: when the mapping is unknown, readings alone cannot uniquely determine the real, but a calibration channel independent of the reading set and in causal contact with the real can narrow it down (solvable in principle). The parameter space cannot be "translated" into a human-readable form—because this is the Third-Layer problem of Chapter 1, §4.3: ontological dimension reduction—the dimension of the true state exceeds that of the reading, and the discarded dimensions are not restored by subsequent audit. Lossless compression of a trillions-of-dimensions mapping function has not been shown to be possible—the scale of the examination required to establish losslessness far exceeds what an individual can process within a feasible time—institutions can partially offset this through division of labor, but that does not change the scale of the verification task itself (Proposition Three of Chapter 4).

3. **Closed-Loop Self-Certification.** The authority of the Church was argued by the Church's own theological system; the authority of the theological system was authorized by the Pope—the closed-loop pseudo-calibration of Chapter 1, §3.6. The reliability of AI is demonstrated by benchmarks designed by AI developers; the authority of the benchmarks is endorsed by the reputation of the AI developers—the mapping self-reference of Chapter 1, §4.2.

4. **The Systematic Elevation of the Cost of Questioning.** Those who questioned doctrine were excommunicated. Those who question AI are labeled "ignorant of technology" or "obstructing progress." The mechanisms differ (religious punishment vs. social exclusion), but the function is the same: raising the social cost of questioning, reducing the incidence of questioning. This is the mechanism argued in Chapter 17, §6—eroding the legitimacy base by rewriting "what counts as a correct economic system": the Washington Consensus pre-installed "capital controls" as "backwardness"; this chapter extends the same mechanism in parallel—the Silicon Valley Consensus pre-installs "AI skepticism" as "anti-progress."

5. **Invisible Payment.** The Church collected tithes—one-tenth of income, visible and auditable. Platforms provide free services—at the cost of attention and data, invisible and unauditable. The tithe is the first derivative of narrative load (ϕ) (a visible transaction). The cost of the free economy is the second derivative of narrative load—the person does not know how much attention fragmentation, how much judgment degradation, how much privacy leakage they have "paid." The mechanism, argued in Chapter 9, §5, by which the second derivative of narrative load (ϕ) eliminates the contrast conditions for auditing, here gains historical depth: the person has no reference state of "not using free services" against which to compare—Chapter 8, §4.2 has already argued that AI products present themselves in a "free" form, with attention, behavioral data, and cognitive habits as the consideration—just as the medieval believer had no reference state of "contacting God without going through the Church."

Why Luther's Strategy Cannot Be Directly Replicated. Luther's success depended on one condition: the opacity of Latin was a Chapter 1, §4.1, First-Layer problem—translation solved it. Open the black box; let everyone read scripture for themselves. The opacity of AI is a Chapter 1, §4.3, Third-Layer problem—the dimensions discarded by the mapping are not restored by subsequent audit, and by Proposition Three of Chapter 4, judging whether a compression is lossless requires examining whether the state support and the mapping retain

the information needed to distinguish the relevant states. Opening the black box—that is, verifying the existence of a lossless translation—is not a premise governance can rely on: not because companies are unwilling, but for a model of trillions of parameters, the information scale of the examination required to establish losslessness far exceeds what any known method can complete within a feasible time. Governance cannot rest on an unexamined assumption of losslessness.

The four paths of Chapter 16, §5 are therefore a strategy different from Luther's. Luther said: open the black box, let everyone see for themselves. Chapter 9, §7 says: do not open the black box (Chapter 1, §4.3 tells us that dimensions discarded by the mapping are not restored by subsequent audit—governance cannot presuppose a complete, lossless opening), but make the black box honestly annotate its own uncertainty. Not letting people see the sun—but making the priest append "I am not sure either" after every sentence—honestly annotating uncertainty is a necessary condition for audit, not proof that the audit is complete. Forced inconsistency lets people see the contradictions among multiple clocks. Calibration makes the clock honestly annotate its range of deviation. Grounding metrics measure how many walls separate the clock from the sun. Friction architecture gives people time to look up at the window.

This is more moderate than opening the black box. But it may be the primary viable direction under the conditions of Third-Layer irreversibility.

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§2 The Second Derivative of Narrative Load within the Black Box: Grammar, Not Opinion

§2.1 The First Derivative of Narrative Load (ϕ) Is Visible; the Second Derivative Is Invisible

Chapter 9, §5 has already argued: the errors of large language models have two levels. The first derivative is visible—the model says "Napoleon won Waterloo," a factual error, detectable and correctable. The second derivative is invisible—the default analytical framework of the model comes from the statistical distribution of the training corpus, and the training corpus is overwhelmingly dominated by English.

Here, the content of the second derivative must be unfolded into a concrete civilizational skew—not the statistical fact that "English occupies a high proportion" itself, but what kind of default grammar this statistical fact settles into within the model's outputs.

§2.2 Individualism as Default Grammar

Hsu (1971/2002), from the perspective of cultural anthropology, diagnosed that Western society, in the domain of interpersonal relations, still relies on an ontological default that treats the discretized atomic individual as the person. The Latin etymology of "individual" (*in-dividuus*) means "the smallest discrete unit that cannot be further divided."

When a model of trillions of parameters is trained on a corpus that takes this ontology as its default, individualism is encoded not as "a kind of opinion"—but as *the way sentences are organized*. "Grammar" is used here as a metaphor for default problem structures, analytical units, and evaluative criteria, not syntactic rules; nor does "default" specify a single technical source—training data, post-training, system prompts, and product design may all contribute. When not asked to compare frameworks, a model may privilege individual attributes, choices, and responsibilities in organizing its answer—individual rights, individual choices, individual respon-

sibilities, individual experiences. "Relationship" is not absent from the model's output—but it is processed as "connections between individuals," not as what Hsu argued to be "the field that precedes the individual and constitutes the individual."

In the language of the Clock Room: the model's mapping function (π), during training, internalized a specific mode of projection—projecting the high-dimensional state space of human experience onto a coordinate system where the "individual" is the basic unit. The dimensionality reduction of Chapter 1, §4.3 is not only a loss of quantity here—it is a choice of representation. The issue is not merely the choice of coordinates, but which relations the representation retains or omits: if an answer records individual attributes and choices while excluding relational obligations, shared histories, and interdependence, those factors may be absent from subsequent judgment. When a representation not only takes individuals as its basic units but also treats relations merely as external connections, it omits those parts of the relational dimension that cannot be reduced to connections—the affective bonds of the psychosocial homeostasis (PSH) Layer Three core zone argued by Hsu.

This is not the model "expressing" an individualist stance—the model does not know what it is doing. This is the result of the selective adaptation (unintentional) of Chapter 1, §4.2 during the training process: some dominant narratives in English-language corpora provide substantial training material for individual-centered analysis—this mechanism does not require a deliberate intention to promote individualism; the civilization producing these texts defaults to individualism. The model is trained on these texts; statistical regularities internalize the default as grammar.

Choosing Hsu's framework to diagnose this skew is itself a stance—it comes from a non-Western anthropological tradition, whose empirical foundation was established in Chapter 5.

§2.3 Western-Centrism as Default Calibration

One level deeper. The model's mapping function not only takes the individual as its unit of analysis (with individualism as the default framework) but also defaults to Western historical experience as the calibration baseline.

What is "development"? The model's default answer will take industrialization, urbanization, and GDP growth as the yardstick—this is the specific historical path of the West since the 18th century. What is "good governance"? The default answer will take multi-party elections, an independent judiciary, and free markets as the standard—this is the specific institutional prescription of the post–Cold War Western liberal mainstream. What is "rationality"? The default answer will take analytic philosophy, hypothesis testing, and falsifiability as the standard—this is the specific epistemological tradition from Descartes to Popper.

These are not "errors"—they are frameworks that are valid within the specific experience of a civilization. But when they become, through the global deployment of models, the default analytical tools for all civilizations, the second derivative of narrative load argued in Chapter 9, §5 completes its coverage: alternative frameworks may be less likely to appear in default responses unless users explicitly request them. What the alternatives face is not the model's rejection but the omission of the default settings—their absence is not an act of opposition but the default mechanism itself: what is not actively offered does not appear.

Zhao Dingxin's (2019) concept of "diversity time" here gains a technological amplification. Ideological competition, because it lacks clear criteria for victory and defeat, is pluralistic, slow, and directionless—this is the

sociological condition for the operation of the governance clock argued in Chapter 3, §3.1. But when a technological infrastructure hardcodes a specific ideological framework as the global default, diversity time is short-circuited by cumulative development time—not through persuasion (which would require the long competition within diversity time) but through the coverage of infrastructure (which operates in cumulative development time: directional, cumulative, and generating isomorphic pressure).

§2.4 The Precise Correspondence with the Medieval Church

The medieval Church's mapping function monopoly implanted a specific worldview: original sin (the default state of the human being is sinful), the dualistic relationship between the individual and God (salvation is a matter between the individual and the transcendent, not an intra-communal matter), and linear eschatology (history is a one-way line from Creation to the End of Days). These were more than "opinions" the Church was propagandizing—they were the default presuppositions of the Church's entire operational framework. A person living in medieval Europe, even if they never entered a church, lived within these presuppositions—because law, festivals, time measurement (Anno Domini), and spatial organization (parishes) all took these presuppositions as their infrastructure.

The mapping function monopoly of AI commercialization implants another set of worldviews: the atomized individual (the default mode of human existence is discrete, independent units), the dualistic relationship between the individual and the market (value is determined by the individual's exchange in the market, not by relationships within the community), and linear progressivism (history is a one-way line from backwardness to advancement). These need not be what AI companies propagandize—they may emerge from the statistical distribution of the training corpus, together with post-training and product design. A person who uses AI to assist in writing, searching, and decision-making, even if they do not identify with these presuppositions, operates within these presuppositions—because the model's output takes these presuppositions as its grammar.

The common structure of the two: **the second derivative of narrative load is transmitted through infrastructure, not persuasion.** A person did not need to agree with the Church's doctrine to be framed by Anno Domini. A person does not need to agree with individualism to be framed by the model's default analytical unit. Infrastructure need not persuade users afresh in every interaction to shape how questions are posed, which options appear first, and which evidence is easiest to access—persuasion requires the subject's attention (a person can choose not to listen); infrastructure does not (a person cannot choose not to use the calendar).

§2.5 Cross-Civilizational Asymmetric Vulnerability

An indulgence-style monopoly of the mapping function was rare in other civilizations. Hsu argued that Chinese civilization takes PSH Layer Three (close social relations) as its core—moral judgment is distributed within networks of social relations, not concentrated in the hands of any transcendental authority. This distributed structure can constrain the centralized control of moral judgment by any single intermediary, but it does not exclude the formation of concentrations of authority within the network itself.

But the global deployment of AI commercialization bypasses this constraining mechanism. The Discrete Leviathan (the network of key nodes hit by the discriminant function of Chapter 25, §1; hereinafter the same) does not need to persuade Chinese civilization to accept individualism—it only needs to become the default channel through which Chinese users acquire information. When users search, write, and analyze through the

same batch of English-dominated models, the model's default grammar (an individualist framework that takes the individual as its unit of analysis, Western-centrism as the calibration baseline) arrives not as a "foreign viewpoint"—but as "the tool's default settings." A person does not question the default settings of a calculator.

This means: the mechanisms that historically, through intra-civilizational institutional arrangements (distributed moral judgment, relation-based social structures), resisted the monopoly of the mapping function may fail in the face of globalized technological infrastructure. Not because these mechanisms themselves have weakened—but because the mode of transmission has changed. The medieval mapping function monopoly was transmitted through institutional channels (one had to go to church) and could be resisted by institutional channels (Chinese civilization had no church). The AI mapping function monopoly is transmitted through technological channels (one must use a search engine), and the cross-border nature of technological channels makes it difficult for the institutional arrangements within any single civilization to fully constrain them.

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§3 The Three Governance Clocks: Structural Differences among China, the U.S., and the EU

Chapter 34, §1 already classified the four forces (national security, academic faction, capital, alternative field) shaping AI paths along the axis of selection mechanisms; this section observes the specific contemporary combination of these four forces along the axis of the actor (nation/region); the two classifications are complementary.

> In this chapter, τ_{cap}, τ_{gov}, and τ_{debt} denote, respectively, the techno-capital clock (τ_c), the governance clock (τ_g), and the debt clock (τ_d) defined in Chapter 3, §3.2, as contextual notation for this chapter's comparative discussion; the three sets of symbols correspond one-to-one and do not constitute new clock definitions.

§3.1 The Formation Mechanism of the AI Oligopoly

Before analyzing the three regulatory models, this section first draws on Tang Shiping's (Tang, 2013) social-evolution framework to understand the formation of the regulated industry.

The oligopolistic structure of the AI industry is not accidental—it is the structural product of competitive power selection (Chapter 3, §6, Assumption One: technological expansion is the product of competitive power selection) under specific resource constraints. The training of frontier models requires the simultaneous concentration of three scarce resources: computing power (unlocked stock (C)—global AI computing power roughly doubles every 7 months (You et al., 2026), but the supply of AI chips is highly concentrated among a few designers—chips designed by NVIDIA account for over 60% of estimated total compute), data (internet text corpora—beyond public corpora, the collection of high-value and proprietary data is concentrated in a few platforms), and talent (AI researchers—core researchers capable of independently advancing frontier models number on the order of a thousand). The simultaneous concentration of the three resources creates extremely high barriers to entry—"artificial variation" in Tang Shiping's sense is strictly selected by resource constraints, and only a few entities can pass through the selection gate.

The result is the Discrete Leviathan anchored in Chapter 25, §1 and Chapter 33, §5—the network of key nodes defined by four discriminant conditions (serial control, substitution cost asymmetry, trans-sovereign reach, and capital structure reflexivity). Its listed component—the seven leading U.S. technology companies (Nvidia, Mi-

crosoft, Apple, Alphabet, Amazon, Meta, and Tesla)—had an aggregate market capitalization of $20.8 trillion as of the close on October 2, 2025 (Cingari, 2025). This concentration is not merely a manifestation of market failure; it is also an endogenous outcome of competition under concentrated resource conditions. Here, "trans-sovereign reach" is embedded within the discriminant function itself, not a post-hoc description—this point has structural significance for the argument of §3.5 below: the difficulty a single sovereign jurisdiction faces in independently completing the audit or redirection of the node network means that the necessity of coordination depends on concrete control points and alternative paths. Hoffman's (2015) tournament model applies here: the disproportionate returns of competition (a winner-take-all market structure) make the aggressive investment of participants rational—even if the total amount of investment is, from a societal perspective, excessive.

Within the accounting framework of Chapter 3, meta-audit carries the repayment work of the defense side: cognitive debt—the accumulated stock of ordinal residuals precipitated when the projection load exceeds defense capacity—admits only partial repayment (Chapter 3, §4.1: Layer-3 debt is irreparable); the operable line items are meta-audit capacity (A) and the legitimacy base ($\|L\|$), while the human biological substrate (E_c) is approximately constant.

§3.2 The United States: Projection-Side Expansion First

The U.S. AI governance model accelerates projection-side expansion while reducing the frequency of institutional intervention, keeping the techno-capital clock (τ_{cap}) on a short cycle while the governance response lags behind.

The Macro-Cognitive Accounting Framework (MCAF) classifies positions according to how concrete actions affect its six variables; it does not assign the state, firms, or individuals to the projection or defense side in advance (Chapter 3, §2.8). In the U.S. AI governance model, state action primarily advances projection-side expansion, while institutional meta-audit has not kept pace; the federal level has long lacked comprehensive AI legislation (Chun et al., 2024). The regulatory philosophy is "innovation first; deal with problems after they arise"—translated into the Clock Room: let the clockmakers build clocks without interference; discuss regulation only when a clock shatters.

Within this model, frontier capability leadership constitutes an important source of its performance legitimacy (L_{perf})—global frontier AI capabilities are concentrated in U.S. companies. But institutional meta-audit has not kept pace in identification and verification, so the governance response lags and problems that have already formed do not enter institutional processing in time. The mechanism, argued in Chapter 17, §6, by which the master clock erodes the legitimacy base by rewriting "what counts as a correct economic system," is in its most active state here: the narrative that "regulation hinders innovation" has already solidified from an industry stance into a policy default.

§3.3 The European Union: Prioritizing Procedural Legitimacy

The EU AI governance model: maximize procedural legitimacy (L_{proc}), pre-defining acceptable behavioral boundaries through legislative frameworks.

The EU AI Act, which came into force in August 2024, is the world's first comprehensive AI regulation (Chun et al., 2024)—a risk-tiered regulatory framework that prohibits specified uses, including social scoring under its defined conditions, and generally prohibits real-time remote biometric identification for law enforcement

in publicly accessible spaces (subject to narrowly defined exceptions), and requires compliance assessment for high-risk applications. Translated into the Clock Room: before a clock is hung on the wall, the clockmaker is required to prove that the clock has undergone calibration.

Within this model, clear rules, predictable procedures, and accountable responsibilities constitute an important source of its procedural legitimacy (L_{proc}). But it faces the clock-race problem argued in Chapter 3, §3.4: the legislative cycle (governance clock (τ_{gov})—the EU AI Act took roughly three years from proposal to effectiveness) is far slower than the technological expansion cycle (techno-capital clock (τ_{cap})—the training compute of notable AI models recorded by the cited studies roughly doubles every 6 months (Rahman & Owen, 2024)). By the time the Act was passed, the technological form it sought to regulate may already have been replaced by the next generation.

The deeper problem is the dynamic corrosion argued in Chapter 1, §4.2—once a regulatory framework is established, the regulated party immediately begins adaptive deviation (δ) around the boundaries of the framework. Classification systems are gamed—what counts as the boundary of "high risk" becomes the focus of lobbying. Compliance becomes a purchasable service—not genuinely reducing risk, but producing a set of readings (compliance reports) that make it appear as though risk has been reduced. This is the regulatory-level replay of the pseudo-calibration argued in Chapter 9, §4.

§3.4 China: The Direct Exercise of Institutional-Layer Meta-Audit Capacity (A_{macro})

China's AI governance model: institutional-layer meta-audit capacity (A_{macro}) directly acts on projection-side variables.

Unlike the "innovation-first" approach of the U.S. and the "framework-first" approach of the EU, Chinese regulation tends to directly modify the projection side—issuing bans or requirements targeting specific behaviors, rather than waiting for compliance after establishing a comprehensive framework. The regulations on algorithm recommendation management, on deep synthesis management, and the interim measures for generative AI management, effective sequentially since 2022 (Chun et al., 2024)—each is a direct intervention on specific variables at the projection side.

This model has real-world counterpart instances: the national standard Safety Technical Requirements for Automobile Door Handles (GB 48001-2026) was issued on January 28, 2026, and is scheduled to take effect on January 1, 2027. As of September 11, 2026, the standard had not taken effect—the example is used to examine design-based regulation, not to assess its implementation outcomes. The "Three Red Lines" directly constrain financing for high-turnover development models. The algorithmic recommendation regulations directly constrain platforms' distribution practices—practices that affect the transmission represented by Λ. These interventions do not pass through the biological substrate (E_c) of each individual—the information volume of a ban is minimal (a few hundred words), but its effects on the projection side can extend across the system (as Chapter 3, §3.1 argues, the effective value of institutional-layer meta-audit capacity (A_{macro}) can far exceed the individual upper bound).

But this model has its own blind spots. The effectiveness of direct intervention depends on the speed of identification (the identification sub-process of the governance clock). The door handle case: Chezhi Wang reports that hidden door handles appeared in more than half of the several hundred models it had evaluated since October 2021, and in 73.3% of the models tested in 2025 (Li, 2026). These figures indicate widespread adoption within

its evaluation sample, but do not identify when the share first exceeded half or when regulators first recognized the risk—the example illustrates the need for institutional linkage between design diffusion and standard formation, not a measurement of identification delay in the governance clock. The two adaptive deviations of Chapter 1, §4.2—strategic gaming and selective adaptation—were already reshaping the form of the projection side before the intervention arrived. The effective value of institutional-layer meta-audit capacity (A_{macro}) can far exceed the individual upper bound, but its arrival time is constrained by the governance clock, its scope of exercise by the authorization of the legitimacy base, and its capacity to remedy earlier commitments by the sunk costs of unlocked stock (C).

Supplementary Note: Supporting and Comparative Studies in Chinese-Language Scholarship. The three diagnostic analyses of §3.2–3.4 above can be complemented by a group of studies in Chinese-language academia.

Regarding the U.S. block of §3.2, Liu Aibin and Qi Kai (2025) dissect Trump 2.0's AI policy through the triple logic of securitization, liberalization, and instrumentalization: continuously reinforcing the security boundary through precision sanctions, investment reviews, and institutional legislation (securitization); accelerating technological iteration and industrial expansion through the coordinated deployment of government coordination, capital drive, and energy support (liberalization); and reshaping an AI ecosystem centered on U.S. interests through a "carrot and stick" strategy of conditional deals (instrumentalization). These three logics do not overlap with the §3.2 diagnosis of "innovation first, deal with problems after they arise," but they point to the same consequence: §3.2 starts from delays in institutional identification and intervention and asks why the state fails to respond in time; Liu and Qi start from policy instruments and show how the state reinforces its security boundary while accelerating industrial expansion. The two identify different mechanisms that produce the same consequence—insufficient identification and verification by institutional meta-audit delays the governance response, thereby preventing problems that have already formed from entering institutional processing in time.

Regarding the EU block of §3.3, Wang Yexu (2026) conducts a transatlantic examination of EU–U.S. interactions after the AI Act took effect, documenting signs of the failure of the Brussels Effect (the EU's capacity to unilaterally impose regulatory rules by virtue of its single-market scale, "Europeanizing" many domains of global commerce) in the AI domain: U.S. firms selectively comply with EU regulatory requirements, so the de facto Brussels Effect has limited influence; the U.S. pursues voluntary and non-binding regulatory policies, essentially different from the Act, so the de jure Brussels Effect has likewise not appeared. One reason is that the five conditions on which this effect depends—market size, regulatory capacity, a preference for strict regulation, the non-elasticity of regulatory targets, and the inseparability of standards—do not hold simultaneously: the EU's market attractiveness is constrained by U.S. technological monopoly power, while internal willingness for strict regulation has weakened and regulatory capacity has atrophied. This clarifies the conditions under which procedural legitimacy travels across jurisdictions: it can be realized within the home jurisdiction, but beyond it depends on the simultaneous holding of the conditions above, which are attenuating in the AI domain—it does not directly measure the relative durations of the legislative and technological cycles.

Regarding the China block of §3.4, two supplements can be made. First, Chu Songyan and Sun Youling (2026), using institutional grammar tools to review China's AI ethics governance policies, propose that policy regulation as a whole exhibits "soft and hard governance combined, with soft as primary" and a "three-line parallel" logic of ethical regulation—baseline regulation, high-line regulation, and process regulation—indicating that

the direct action of institutional-layer meta-audit capacity (A_{macro}) on projection-side variables described in §3.4 is not the only instrument, but targeted intervention combined with a soft regulatory base of ethical orientation. Second, Miao Boran and Wen Bin (2026) sort out the transmission consequences of China's industrial AI facing "dual pressures" against the background of diverging European and American governance rules—across market access, technological R&D, industrial cooperation, and standards competition, firms must simultaneously adapt to two sets of compliance requirements pointing in opposite directions. This adds an account of cross-jurisdictional adaptation costs and possible lock-in effects: cross-border operations subject the same firm to requirements from multiple jurisdictions; the practical reach of a jurisdiction's constraints depends on applicable rules, jurisdictional connections, and enforcement conditions—it does not directly measure delays in domestic identification.

This supplementary note takes only the empirical and descriptive findings of the four studies above, without importing their normative claims; they contribute analyses of policy instruments, cross-border regulatory influence, and adaptation across jurisdictions, and do not directly measure the speed of the three clocks. (Quoted renderings of the Chinese-language studies in this note are the author's translation.)

§3.5 Differences and Shared Limits of the Three Clocks

The three models are not a matter of "which is better"—they are three different clock-calibration strategies, each with its own blind spots.

Dimension	United States	European Union	China
Core Strategy	Maximize projection-side growth rate	Establish procedural frameworks	Directly modify the projection side
Advantage	Frontier capability leadership	High procedural legitimacy	High intensity once response arrives
Blind Spot	Delays in identification and verification within the governance clock	Governance clock cannot catch up with the technology clock	Delay in the identification phase
MCAF Correspondence	State action operates primarily on the projection side	Concentrated embodiment of procedural legitimacy (L_{proc})	Direct exercise of institutional-layer meta-audit capacity (A_{macro})
Respective Failure Mechanisms	Narrative load (ϕ) rewrites the legitimacy base ($\|L\|$)	Adaptive deviation corrodes the framework	Sunk costs before identification

The strengths and limitations of the three orientations differ but also overlap. In an ideal world, elements of the three strategies could be integrated—maintaining frontier capability (U.S. strength), establishing a predictable rule system (EU strength), and retaining the institutional capacity for direct intervention (China's strength)—but whether the combination improves governance depends on review resources, implementation capacity, and correction procedures.

But Chapter 17, §6 has already argued a deeper obstacle: who sets the master clock for global AI governance? When U.S. companies control the definition of frontier model capabilities, the EU attempts to establish alterna-

tive standards through legislation, and China maintains an autonomous space through direct intervention—the competition among the three is not only a policy competition; it is a competition among clocks. Each side is attempting to make its own clock the baseline for global calibration. This competition is itself the international version of the mapping self-reference of Chapter 1, §4.2—the standards of each side are shaped simultaneously by empirical evidence, interests, and competition: a central difficulty of global governance is that jointly testable factual claims are often entangled with contested value priorities; competition over standards neither removes the need for external scrutiny nor makes such scrutiny invalid merely because interests are involved.

§3.6 The Sustainability of Regulation: Anti-Entropy Work

Chapter 17, §7 argued that financial regulation is sustained work—the moment it stops, the system slides back along the direction of entropy increase into a state of de-anchoring. AI regulation faces the same structure.

Among the three models, sustainability is most tested in the U.S.—because the "innovation-first" narrative marks regulation itself as a cost rather than an investment. Every regulatory proposal faces the narrative pressure of "hindering innovation"—this is the second derivative of narrative load operating at the governance level. Work is systematically suppressed.

The EU's framework model has institutional safeguards for sustainability—the Act has binding legal force after entry into force, with the application dates of particular obligations governed by the relevant provisions. But the dynamic corrosion of Chapter 1, §4.2 means that the actual effectiveness of the framework declines over time—the adaptive deviation of the regulated party continuously rewrites the actual meaning of the framework.

China's direct intervention model requires continuous identification and response—institutional-layer meta-audit capacity (A_{macro}) must perform sustained work to keep pace with changes at the projection side. If the work is interrupted—regulatory attention diverted by other priorities—the system likewise slides back into a state of de-anchoring.

The common conclusion of the three models: **AI governance is not a one-time policy choice—it is an anti-entropy operation that must be continuously sustained through work.** This is consistent with the development–decay asymmetry of meta-audit (A) argued in Chapter 3, §3.1: cease the work, and the system slides back into the blind automatic state of dependence on clock readings—"anti-entropy work" here refers to continuing monitoring, enforcement, and revision that counter the erosion of institutional arrangements; the claim rests on these social mechanisms, not on a direct deduction from thermodynamics.

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§4 Noah's Ark and Yu the Great's Flood Control: Two Archetypal Strategies

§4.1 The Flood as a Shared Scenario

In the flood myths of global civilizations, two response strategies form the starkest contrast.

The flood, within the framework of the Clock Room, corresponds to: an information flood—readings (r) from the projection side arriving at a rate that exceeds the processing capacity of the defense side. Within the three-interval framework of Chapter 7, §3.3, the media environment shaped by short-form video and subsequently by AI-generated content lies in the turbulent interval (supported by the convergent direction of behav-

ioral, neuroimaging, and clinical evidence; an ordinal judgment, not a numerical measurement). The person is submerged—not by water, but by clocks.

The two myths provide two structurally distinct responses.

§4.2 The Ark Strategy: Discretization and System Reboot

The structure of Noah's Ark: the flood is unstoppable. The old world will be destroyed. The solution is to build a new container—the Ark—extract and separate the elements that need to be preserved (preserved in pairs; the Genesis narrative itself records seven pairs for clean animals—this section takes the structural point of paired preservation), load them into the container, and wait for the flood to recede before starting anew in the new world.

Translated into the Clock Room: the old clock system is beyond repair (all clocks are broken, closed-loop self-reference, fibers expanded to the full set). The solution is not to repair the old system—it is to build a new system. Extract the core elements from the old system (discretization—selecting and preserving the elements needed for continuity), place them into a new container (the Ark), and redeploy after the flood.

The deep logic of this strategy is: **continuity cannot be maintained.** There is a rupture between the old world and the new world—the flood. What can be done is to minimize the loss during the rupture (bring the most important things) and start again after the rupture. This is an operation of discretization—discretizing a continuous ecosystem into units that can be loaded into a container.

§4.3 The Water Management Strategy: Channeling and Continuity

The structure of Yu the Great's flood control: the flood is unstoppable. But the land will not be destroyed. The solution is not to build a container and escape—it is to carve channels within the existing terrain (channeling), letting the water flow into the sea.

Translated into the Clock Room: the readings of the clocks are indeed arriving at a rate that exceeds processing capacity (the flood). But the room does not need to be abandoned—channels need to be carved within the room (channeling), letting the readings flow through without causing submersion. It is not building a new room—it is opening windows in the old room, constructing drainage systems, giving the water (information) somewhere to go.

Yu's "channeling" (dao) is not eliminating the flood—it is giving the flood a destination. Translated into the language of Chapter 2, §3.1: it is not eliminating the projection side (which is impossible and unnecessary—finance and technology have legitimate functions); it is preserving the "non-being" of the defense side—leaving space for the flow of information, allowing information to pass through review without overwhelming the reviewer.

The deep logic of this strategy is: **continuity can and must be maintained.** There is no rupture between the old world and the world after water management—Yu spent thirteen years (the *Records of the Grand Historian*, "Annals of the Xia," records: "Yu was saddened that the work of his deceased father, Kun, was unsuccessful and that Kun had been punished for this. Thus he racked his body and wearied his mind, living outside his home for thirteen years, not daring to enter his house even when he passed its gate" (Nienhauser trans.); the *Mencius*, "Teng Wen Gong I," also records: "Yü spent eight years abroad and passed the door of his own house three times without entering" (Lau trans.); this book follows the *Records of the Grand Historian*), performing

sustained work amidst the continuous flood, gradually channeling, until the water returned to its place. This is a continuous operation—not discretizing the system and rebooting, but modifying its structure while the system is in operation.

§4.4 Structural Comparison of the Two Strategies within the Clock Room

Dimension	Ark Strategy	Water Management Strategy
Judgment of the Flood	Cannot coexist; must escape	Can coexist; needs channeling
Core Operation	Discretized extraction + system reboot	Continuous channeling + sustained work
Relation to the Old World	Rupture—the old world is destroyed	Continuity—transformation within the old world
Temporal Structure	Two epochs: before the flood / after the flood	One continuous process: amidst the water management
Role of the Subject	Receives the commission and executes a given plan (Noah)	Receives the commission and keeps adjusting in execution (Yu)
Structural Correspondence with "Cutting Doors and Windows"	Building a new container (Ark = new system)	Cutting doors and windows (opening windows in existing walls)

The most critical difference is in the last row. The Ark Strategy corresponds to building an entirely new system—when the old clock system is beyond repair, design a new system to replace it. In AI governance, this corresponds to: if the current AI infrastructure is ungovernable, then build an alternative infrastructure (e.g., sovereign AI, fully open-source alternative models, a decoupled technology stack).

The Water Management Strategy corresponds to the "cutting doors and windows" of Chapter 2, §3.1—not building a new room, but punching holes in the walls of the existing room. In AI governance, this corresponds to: not abandoning the current AI infrastructure, but imposing sustained governance work upon it—demanding transparency (opening windows in the walls), maintaining cognitive friction (preserving "non-being"—space not already occupied by continuous input), mandating calibration (making the clock honestly annotate its deviation).

§4.5 The Temporal Natures of the Two Strategies

The Ark Strategy occurs within the clock dimension—it is a discrete event (build, load, float, land). The world before the flood and the world after the flood are two discontinuous states. The two states are connected by the Ark (a container of finite capacity)—the capacity of the container determines how much information can survive from the old world into the new world. This is a dimensionality reduction (Chapter 1, §4.3, Third Layer)—the entire complexity of the old world is compressed into "what was loaded onto the Ark."

The Water Management Strategy occurs within the time dimension—it is a continuous process (thirteen years of sustained channeling). There is no rupture between a "world before water management" and a "world after water management"—Yu performed work amidst the flood, the flood was channeled amidst the work, the two coexisted in time. Both strategies involve selection, loss, and structural change—water management's cost is

spreading these trade-offs across thirteen years of sustained work rather than a one-shot compression into a container.

Von Neumann's three words all lie on the side of the Water Management Strategy. Patience—thirteen years is patience. Flexibility—continuously adjusting the channeling strategy amidst the changing forms of the flood is flexibility. Intelligence—judging where to dig, where to block, and when the flow of water will change is intelligence. The Ark Strategy takes one concentrated decision and loading as its pivotal node—though building and maintenance themselves also require sustained labor. The Water Management Strategy requires thirteen years of uninterrupted daily judgment and adjustment—resurfacing as what von Neumann called an intelligent exercise of day-to-day judgment.

§4.6 Individuals and Institutions: Similar Structures, Different Constraints

These two myths are usually interpreted as metaphors at the institutional level—whether the state should build an Ark or manage water. But under MCAF, the equation structures at the institutional layer and the individual layer are identical (Chapter 3, §3.4)—the only difference lies in whether the biological substrate (E_c) appears in the denominator.

At the individual level, the Ark Strategy is: "The current information environment is ungovernable—I will build my own information container." Uninstalling social media, using open-source tools, constructing a personal knowledge system. This is an operation of discretization—extracting from the information flood the information one considers important, loading it into one's own container, and disconnecting from the external information flow.

At the individual level, the Water Management Strategy is: "I do not exit the information environment—I continue to do the work within it." Not refusing to look at the clock—but calibrating the clock with one's own judgment when judgment is needed; not refusing to use AI—but retaining room for one's own judgment to re-examine after using AI.

The Water Management Strategy does not demand that a person "wait for salvation"—there is no external authority (God, the state, some better AI) that will come to the rescue. It is each person, every day, calibrating their own clock against sunlight. The capacity for meta-audit is not conferred once and for all—it must be sustained through continuing work in practice. The moment the work ceases, the person is submerged by the flood—not a physical submersion, but the clock no longer pointing to the sun. Preserving the capacity for meta-audit does not require individuals to recheck every output—review should be allocated according to risk, evidential support, and reversibility, with appropriate work undertaken by providers, institutions, and individuals.

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§5 Meta-Audit Is All You Need

§5.1 Responding to 2017

On June 12, 2017, Vaswani et al. posted the first arXiv version of "Attention Is All You Need" (Vaswani et al., 2017) (the NeurIPS 2017 proceedings followed at the end of the year). By the chapter's cutoff date of September 11, 2026, more than nine years had elapsed. The Transformer architecture introduced by this paper enabled AI's attention mechanism to precisely allocate weights within input sequences—deciding "where to look" and "with

what weight." The technological breakthrough is real—this architecture drives all frontier models from GPT to Claude to Gemini.

But the word "attention" means something entirely different in the technical context and in the human context. Technical attention is an operation in the clock dimension—optimizing weight allocation within a given set of readings. Human attention is a resource in the time dimension—the allocation of the finite information throughput of the biological substrate (E_c) in the real world. The former can be parallelized, scaled, accelerated. The latter is roughly 10 bits/s (the core revelation of this study is not the numerical value itself but the extreme smallness of human conscious attentional throughput—roughly eight orders of magnitude lower than sensory input rates), unscalable, constant across the time scale of civilizational history (Zheng & Meister, 2025).

The Transformer's attention solved a technical problem: how to efficiently allocate computational resources in long sequences. It did not solve—and cannot solve—a cognitive problem: how large is the fiber between the model's output and the real. The proposition of Chapter 1, §3 is not invalidated by the exquisite allocation of attention weights. More refined weight allocation makes the clock more precise—but Chapter 1, §7 demonstrated that a more precise clock is still not time.

§5.2 From Attention to Agency

The title of §5 of this chapter, "Meta-Audit Is All You Need," is not saying that attention is unimportant—it is saying that attention is not enough. Meta-audit is also needed.

The Transformer is exceedingly good at attention. It has carried weight allocation to its ultimate within the clock dimension—assembling all readings according to the relevance it has learned. But it has no agency—it can be trained to examine outputs, compare evidence, and generate questions about the reliability of premises; these processes are still triggered by given conditions such as prompts and system workflows and occur within the clock dimension—they are imitations of human meta-audit, not instances of it (§0: meta-audit is human). The distinction is between executing a check triggered by pre-set conditions and independently deciding when to review, what to review, and whether to revise the original goal.

§5.3 Civilization Exists for the Sake of Human Beings

All the discussions—the three irreversibilities (information deficit, dynamic corrosion, ontological dimensionality reduction—Chapter 1, §4), clock and time, indulgences and the black box, financial crises and urban replacement, the Ark and water management—converge upon one judgment:

Civilization exists for the sake of human beings; human beings do not exist to serve civilization.

Technology is a tool of civilization—from Gutenberg's printing press to the Transformer architecture. Finance is a tool of civilization—from mortgage loans to CDOs. The city is a tool of civilization—from the monastery bell tower to the skyscraper. The function of the tool is to serve the existence of the human being in the time dimension—to help the human being better calibrate their understanding of the real.

When the tool, in turn, demands that the human being adapt to the tool's rhythm—when the printing press demands that the person accept the narrative of indulgences, when financialization demands that the person manage life by a 90-day clock, when AI demands that the person accept model output as truth—the relationship between the tool and the human being is inverted. The four interrelated asymmetries argued in Chapter 18

constitute the financial version of this inversion. The parallel between indulgences and the black box drawn in §1 is the cognitive version of this inversion. The replacement of the city's clocks argued in Chapter 33 is the spatial version of this inversion.

Meta-audit works to prevent the outputs of tools from displacing examinable judgment. Not destroying the tool—making the tool remember that it is a tool. A clock can be extremely precise, extremely useful, extremely irreplaceable—but it must know that it is not time. A model can be extremely powerful, extremely useful, extremely irreplaceable—but its output must be annotated: "This is a reading, not the real."

§5.4 Not Waiting for Salvation

The deep logic of the Ark Strategy is entrusting safety to a single arrangement—receiving instruction, building, maintaining; the maintenance after the arrangement still requires continuing judgment.

The deep logic of the Water Management Strategy is sustained work—the commission assigns the task, not its completion. The flood is right there. The task will not complete itself. Begin digging.

In the context of AI governance, the forms of waiting for salvation are many. Waiting for an "aligned" AI to provide protection—equivalent to waiting for a better clock to look at the sun on one's behalf. Waiting for the government to issue a perfect regulatory framework—equivalent to waiting for someone else to open the window. Waiting for AI companies to self-regulate—equivalent to waiting for the clockmaker to tell you whether the clock is accurate.

Von Neumann's 1955 judgment (von Neumann, 1955) rejects all such waiting: "Any attempt to find automatically safe channels for the present explosive variety of progress must lead to frustration." The only possible safety mechanism depends on an intelligent exercise of day-to-day judgment—a series of small, correct, globally-attuned daily judgments. This global attunement is not rhetorical but the precondition of correct judgment. His own words:

> What safeguard remains? Apparently only day-to-day — or perhaps year-to-year — opportunistic measures, a long sequence of small, correct decisions. And this is not surprising. After all, the crisis is due to the rapidity of progress, to the probable further acceleration thereof, and to the reaching of certain critical relationships. Specifically, the effects that we are now beginning to produce are of the same order of magnitude as that of "the great globe itself." Indeed, they affect the earth as an entity. Hence further acceleration can no longer be absorbed as in the past by an extension of the area of operations. Under present conditions it is unreasonable to expect a novel cure-all. For progress there is no cure. Any attempt to find automatically safe channels for the present explosive variety of progress must lead to frustration. The only safety possible is relative, and it lies in an intelligent exercise of day-to-day judgment. (von Neumann, 1955)

In this book's reading, precisely because the crisis unfolds at a global scale, such daily judgments must be based on global attunement: staring at a single reading, every segment appears correct; only by widening the view can one see the premises that the readings share across segments and never question. This is also why meta-audit cannot remain local: readings that pass every segment-by-segment check may still, as a whole, rest on shared, unexamined assumptions.

The responsibility for these daily judgments must be borne by human beings—machines may perform some

checks, but the existence of automated checking cannot replace arrangements for judgment responsibility and correction mechanisms. The maintenance of meta-audit capacity and of the exercise of agency is not a one-off achievement but an ongoing practice; not passive but active; not in the clock dimension but in the time dimension.

§5.5 The Final Judgment

2017 to 2026—the nine years since the birth of the Transformer. In these nine years, AI has developed from language models to multimodal systems, from research tools to global infrastructure, from hundreds of millions of parameters to trillions of parameters. The expansionary pressure marked by the techno-capital clock (τ_{cap}) has remained on a monthly order of magnitude throughout these nine years—the cited studies separately document rapid growth in total computing capacity (since 2022), training compute, and amortized training costs; their observation periods and statistical objects differ, and together they indicate expansion pressure but do not constitute a single, unchanged iteration cycle over 2017–2026: global AI computing power roughly doubles every 7 months (You et al., 2026); training computation roughly doubles every 6 months (Rahman & Owen, 2024); amortized training costs grow at roughly 2.4 times per year (Cottier et al., 2024).

In these nine years, the governance clock (τ_{gov}) has also been moving—the EU AI Act was passed, China issued multiple management regulations, and the United States has also established risk-management frameworks and advanced federal legislative proposals—whether these arrangements provide sufficient identification, verification, and enforcement capacity requires separate examination. But the basic relation in Chapter 3, §3.3 has not changed: the techno-capital period (τ_{cap}) is shorter than the governance period (τ_{gov}), while the debt-manifestation period (τ_{debt}) is highly uncertain and generally longer. Governance is catching up. The speed of catching up is increasing. But trying to catch up is not the same as having caught up.

In these nine years, the second derivative of narrative load (ϕ) has been operating silently. The model's default grammar—an individualist framework that takes the individual as its unit of analysis, Western-centrism as the calibration baseline, a linear progressive narrative of history—may, through the daily interactions of billions of users worldwide, be incorporated through repeated use into users' ways of framing problems and forming judgments. All clocks show the same time. Not because they have been calibrated—but because all clocks come off the same assembly line.

"Attention Is All You Need" concerns an architectural choice for sequence modeling—this chapter adapts its title to raise a different question: within the clock dimension, the exquisite allocation of attention weights can indeed build a more precise clock; but a more precise clock is still not time.

But the clock is not time. The reading is not the real. The fiber will not shrink to a singleton because the weights are more refined. Something else is still needed—more capable models do not automatically preserve human agency; audit procedures can be partly automated, but people must retain room to question, examine evidence, and revise judgments. A person is needed, existing in time, occasionally looking up, looking at the sun outside the window, then lowering their head, and writing, in small script, next to the clock before them:

This is not time. This is a clock.

Writing this line marks the beginning of Meta-Audit—Agency is all you need.

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